

TUAL DUMATUBUN, Indonesia

WMO: 978100

Lat: 5.6636S

Lon: 132.7361E

Elev: 39

StdP: 14.68

Time Zone: 9.00 (E09)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	70.8	72.6	67.8	102.4	76.4	69.0	106.7	77.2	25.4	79.3	23.0	79.5	0.9	150	0.375

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11	12.8	90.6	80.4	89.7	80.2	88.9	80.0	82.6	87.9	81.9	87.2	81.3	86.6	9.8	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
81.0	161.0	86.1	80.3	157.7	85.5	79.8	154.9	84.9	46.4	88.1	45.7	87.4	45.1	86.8	86.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.2	18.0	16.0	DB	61.5	94.0	4.4	3.8	58.4	96.8	55.8	99.0	53.3	101.2	50.1	103.9	(4)
(5)			WB	62.4	83.7	3.5	1.4	59.9	84.7	57.8	85.5	55.8	86.3	53.3	87.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	81.1	81.4	81.2	81.1	81.4	81.5	80.4	79.7	79.4	80.6	82.1	82.4	81.5	(6)	
	DBStd	2.06	1.87	1.80	2.02	1.85	1.99	1.88	1.50	2.07	1.98	2.12	1.78	1.57	(7)	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)	
	CDD50	11336	974	872	964	942	977	913	922	911	919	996	971	975	(10)	
	CDD65	5861	509	452	499	492	512	463	457	446	469	531	521	510	(11)	
	CDH74	59119	5290	4608	5060	5051	5460	4406	3959	3876	4553	5784	5769	5303	(12)	
(13)	CDH80	16988	1581	1299	1491	1554	1619	943	620	710	1239	2089	2193	1650	(13)	
(14)	Wind	WSAvg	6.6	6.7	6.2	5.2	4.5	6.1	8.4	9.9	9.8	7.7	5.9	4.1	4.6	(14)
(15) Precipitation	PrecAvg	103.2	16.5	13.2	14.8	11.4	10.5	6.0	3.3	2.7	2.5	3.2	6.9	12.2	(15)	
	PrecMax	163.4	24.8	27.6	22.8	22.7	28.1	15.6	10.9	9.9	9.1	6.7	12.9	21.7	(16)	
	PrecMin	64.1	6.9	6.7	7.4	1.0	2.5	0.0	0.0	0.0	0.0	0.2	1.9	5.2	(17)	
	PrecStd	26.6	5.0	5.9	4.8	6.1	6.0	4.1	3.0	2.9	2.2	1.9	3.7	5.5	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	89.8	89.0	90.0	90.3	89.6	88.0	86.4	87.0	89.3	91.6	92.3	90.7	(19)	
		MCWB	80.7	79.9	81.0	80.9	81.2	79.9	78.6	77.9	79.1	80.7	80.4	80.4	(20)	
	2%	DB	88.6	88.0	88.6	89.0	88.4	86.9	85.2	85.6	88.0	90.2	90.8	89.1	(21)	
		MCWB	80.5	79.9	80.4	80.4	80.6	79.3	77.8	77.1	78.3	80.0	80.2	80.4	(22)	
	5%	DB	87.5	87.1	87.5	87.9	87.4	85.9	84.3	84.7	86.8	89.2	89.6	88.0	(23)	
		MCWB	80.2	79.8	80.0	80.1	80.2	78.8	77.2	76.6	77.7	79.5	80.0	80.2	(24)	
	10%	DB	86.3	86.1	86.4	86.8	86.4	84.8	83.3	83.8	85.6	88.1	88.4	86.8	(25)	
		MCWB	79.9	79.6	79.8	79.8	79.8	78.3	76.8	76.2	77.2	79.0	79.7	80.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	82.6	82.5	82.6	82.8	82.9	81.0	80.0	79.5	80.7	83.2	82.8	82.4	(27)	
		MCDB	87.6	86.4	87.6	88.2	87.5	86.2	84.6	84.9	87.7	89.5	89.0	87.6	(28)	
	2%	WB	81.4	81.4	81.6	81.7	81.9	80.0	79.0	78.4	79.5	81.7	81.8	81.5	(29)	
		MCDB	86.5	85.6	86.7	87.0	86.7	85.4	83.7	83.9	86.2	88.4	88.2	86.7	(30)	
	5%	WB	80.7	80.7	80.8	80.9	81.0	79.4	78.1	77.6	78.8	80.5	81.0	80.9	(31)	
		MCDB	85.8	85.1	85.8	86.1	85.9	84.5	82.7	82.8	85.1	87.1	87.3	86.1	(32)	
	10%	WB	80.1	80.0	80.1	80.2	80.2	78.8	77.4	76.9	78.0	79.7	80.3	80.3	(33)	
		MCDB	84.8	84.4	84.9	85.3	85.1	83.6	81.6	82.0	83.8	85.9	86.4	85.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.4	9.5	10.1	10.5	9.8	8.1	7.4	9.1	10.8	12.4	12.8	10.6	(35)	
		MCDBR	11.1	11.3	11.8	12.2	11.2	9.6	8.5	10.1	12.0	13.3	14.4	12.8	(36)	
	5% WB	MCWBR	4.8	4.8	5.3	5.4	5.1	4.4	4.1	4.3	4.8	5.3	5.8	5.6	(37)	
		MCWBR	10.3	9.9	11.0	11.3	10.6	9.0	8.1	9.1	11.4	12.1	13.0	11.6	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.415	0.414	0.402	0.405	0.434	0.440	0.429	0.433	0.446	0.467	0.449	0.419	(40)	
		taud	2.525	2.520	2.563	2.563	2.474	2.444	2.476	2.459	2.408	2.326	2.392	2.496	(41)	
	Ebn at Noon	Ebn at Noon	293	293	292	283	266	260	265	271	275	274	281	291	(42)	
		Edh at Noon	35	36	34	32	34	34	34	36	39	43	40	36	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1511	1611	1607	1585	1501	1380	1471	1676	1842	1949	1885	1584	(44)	
	RadStd	RadStd	182	153	121	107	130	133	115	140	117	135	128	156	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page